

**Improving the administrative decision-making of organizations in Iraq using
information security**

تحسين عملية اتخاذ القرارات الإداري للمنظمات في العراق باستخدام أمن المعلومات

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Abstract:

Information security has become one of the most important aspects of modern electronic society. It is a vital concern and vital issue. The different ways of storing and transfer information in different ways between the network, security threats and challenges have become more complex than ever before. This paper attempts to demonstrate the impact of adopting information security on the administrative decision in the organizations, by proving its impact on the business process, marketing edge, decrease expenses and applying compliance to avoid effects in the near future, as well as to clarify the nature of the threats posed by modern communication technologies, and their applications to the information security system.

The most of researchers studied the impact of information security on organization without shedding light on its importance in the administrative decision-making of the organizations. The researcher relied on surveys of the direct and indirect influence factors on the administrative decision in terms of information security and the most important problems, obstacles and solutions in the giant companies. The researcher outcome of these factors coincided with the administrative decision-making through the work of these Model effects.

الخلاصة:

أصبح أمن المعلومات أحد أهم جوانب المجتمع الإلكتروني الحديث. إنه شاغل حيوي وقضية حيوية. أصبحت الطرق المختلفة لتخزين ونقل المعلومات بطرق مختلفة بين الشبكة والتهديدات والتحديات الأمنية أكثر تعقيداً من أي وقت مضى. تحاول هذه الورقة إثبات تأثير تبني أمن المعلومات على القرار الإداري في المنظمات ، من خلال إثبات تأثيرها على العملية التجارية ، التسويق ، تخفيض النفقات وتطبيق الامتثال لتجنب الآثار في المستقبل القريب ، وكذلك توضيح طبيعة التهديدات التي تطرحها تقنيات الاتصالات الحديثة ، وتطبيقاتها على نظام أمن المعلومات.

درس معظم الباحثين تأثير أمن المعلومات على المنظمة دون إلقاء الضوء على أهميتها في عملية اتخاذ القرارات الإدارية للمنظمات. اعتمد الباحث على استطلاعات عوامل التأثير المباشر وغير المباشر على القرار الإداري من حيث أمن المعلومات وأهم المشكلات والعقبات والحلول في الشركات العملاقة. تزامنت نتائج الباحث لهذه العوامل مع اتخاذ القرارات الإدارية من خلال عمل هذه الآثار النموذجية.

Keywords: Organization, Information security, GDPR, ISO27000, decision-making

1. Introduction

Information security is necessary in organizations, because of the increase in the volume of data and electronic communications on a daily basis[1]. In this case, security attacks have become more complex, making it very dangerous for organizations as it leads to interrupting their daily work and damaging their services, which directly and negatively affects their reputation and thus their profitability[2].

Information warfare has become a fact of the times and a tool of conflict[3]. If you want to the country to be paralysis, it must strike its information system[4]. The primary aim of the hidden or declared conflict is the information system and its infrastructure. The formation of electronic armies has become a reality for decades in developed countries[5].

There are still organizations that are running by Bureaucratic mindsets that do not understand the requirements of the electronic era. It is bad to support and adopt some projects than reluctant that guarantee the security of electronic information and ensure its efficient and sustainable flow[6].

Information technologies are only a part of the information system. Therefore, IT security management is not enough to protect the information system [7]. The proposed article researches the impact of information security on the course of decision-making regarding the best practices of organizations. This research relies on preliminary studies on the expression of factors that directly and indirectly affect the administrative decision and the most prominent problems, obstacles, and solutions in giant companies, which rely primarily on the security of information in underlining its market power. Additionally, this research tries to compare them in the current reality of Iraqi

2. Theoretical background

2.1. Basic principles

Protection and encryption scientists have identified several important elements over the past twenty years for information security organizations that must be identified to be safe[8]. They have known three elements that collectively constitute a triangle called the CIA Triangle of protection as shown in figure 1 [9]. First, Confidentiality, which means that information is protected from unauthorized persons to access information. The credit card is one of the most highly confidential systems, but it is nevertheless violated and stolen. The confidentiality of credit cards lies in the encryption of the card number. Second Integrity, Which means that the information we protect should be correct and non-flawed information, including data protection from any update that may occur from third parties that may not be permitted. Third availability, any system using its own IS, it is necessary to provide the data immediately when the user ask for it and need it. The information should be available when the user wants to access it, and that it is not obscured when needed. Maintaining the Integrity of the Specifications[10].



Figure 1 [9] The elements of the CIA

2.2. Information protection measures

There are three standards to protect information which are Access, Authentication and Auditing. Access is to control through which different methods are used to prevent people from accessing the network or devices .Authentication is to ensure the validity of the users of the system or the information. Lastly, Auditing which ensures the continuous integrity of information and that it is not exposed to the penetration and discovery of hackers at any time .All standards are symbolized by the symbol AAA and are used every day for protection[11].

Classifying protection tools from violations regarding the time of their implementation brings us to being “Preventive” or “Detective”. In the preventive classification, you try to avoid accidents, whilst in the detective classification you aim to detect accidents after they have occurred[12].

Protection type	Preventive	Detective
Administrative (Regulatory procedures)	• Training and security enhancement programs • Separation of tasks • Emergency plans	• Performance evaluation • Rotation of duties and tasks • Security audits and audits
Technical tools (software)	• Smart cards • Virus control • Access control settings • Encryption	• Intrusion screening systems • Tracking access
Physical protection	• Physical location selection • vital means of biological • Alternative power source	• Fire and smoke detectors • Sirens and sensors

Table [1] Shows examples of preventive and detective protection[12].

Prof Han mentioned cloud computing and virtual data infrastructure are applied information system as the technology, which sees itself as technology economic cost. For example, robotic, artificial intelligence and big data are more expensive to implement in organizations [13]. The organization should develop a robust data mining system so to be able to take advantage of it at the management level, and be able to make decisions with sufficient information[14].

According to a recent Gartner study, governments have been slow to strengthen and expand these initiatives, compared to other sectors .The study indicated that half of the participants from the government sector are looking forward to the transition to digital governance in order to achieve a combination of the goals of digital transformation and improvement of operations .The other half focuses on only one objective: improving operations (33 %) or achieving digital transformation (17 %) [15].

Governmental organizations are seeking to expand their digital workforce by collaborating with partners, including employees, citizens, consumers, startups, digital giants and service providers, which contribute to enhancing the benefits and benefits of transforming the digital government[15].

The study organizations in Saudi Arabia confirmed that there are two sources that negatively affect the security of information. External security threats, and threats within the organization (internal security). Organizations willing to reach an adequate level of security must be able to identify security holes and develop a mechanism to prevent any misuse of them[16].

Any organization that aims to achieve information security management must include the following components[17]:

- Physical security: including sources, property, and buildings to prevent illegal access
- Individual security: to protect individuals and groups with access to information
- Security operations: to protect the activities and operations carried out by the authorized
- Communication security: to protect the media and technology used and the content
- Network security: to protect network components, messages, and contents

- Data security: To protect the confidentiality, integrity and availability of information. Most of the organizations fall victim to such attacks because of weak information security policies (ISPs). For this reason, organizations are under information security threats. Therefore, any useless of IS policies by IT users are putting the organization at risk[18].

As mentioned in Maryam's research, there are five deterrent methods of a punishment which deters employees from violating the information security policies of the organization. These consist of personal insult, insulting the family, defeat the work, penalty or imprisonment for financial loss, one's conscience[16].

In the same context, there is a study conducted on 346 employees working in several organizations. Therefore, whether employees' security-related stress, technostress, and role stress, all of these reasons that effect in an organizational setting could affect their compliance intention regarding information security. The results reveal that they were found to adversely affect the organizational commitment of employees when pushing them both directly and indirectly under the stress, and further lowered compliance intention. For this reason, the researcher found that managing and enhancing the employees' information security compliance is very important to implement in organizations through organizational strategies[19].

3. Information Security Factors in Organizations

3.1. Compliance

Compliance policy is one of the most important factors affecting the work of organizations and companies in terms of security[20]. It ensures that all processes and procedures comply with international standards, such as maintaining the confidentiality of customer information and complying with secure and reliable transactions in the exchange of information and data, thereby enhancing customer satisfaction and risk management[21]. It handles most common threats from device threat reports. Contributed a group of Internet security experts, including the most famous organizations in response to incidents and crimes, in the development of these procedures, and controls. In addition, actions are constantly updated in response to increased threats and attacks.

- GDPR: GDPR is a summary for General Data Protection Regulation [22]. GDPR works to harmonize data privacy laws across Europe to protect and enable the privacy of EU citizen's data in the EU leading to restructuring the way organizations in the region deal with data privacy [22]. Most sources suggest that GDPR is a defining moment in defining new privacy standards worldwide[23].
- ISO 27001: This standard provides a periodic model known as the PDCA (Plan-DO-Check-Act), which aims to identify requirements for establishing, implementing, operating, monitoring, reviewing, maintaining, improving and documenting the information security management system within the organization. It usually applies to all types of organizations, including businesses, government agencies, and others[24].
- COBIT: refers to Control Objectives for Information and Technology is a control or control framework that links information technology to business requirements, organizes IT activities into the accepted process model, identifies key IT resources, and controls administrative objectives to be considered. This standard was built by the IT Governance Institute (ITIG) in 1995[25].
- Standard ITIL: The Information Technology Infrastructure Library is also called the ISO 20000. It is a collection of best practices in the management of IT services (ITSM) and focuses on the IT service operations and the role of the user. It was built by the United Kingdom Office of Government Trade (OGC)[26].

- The PCI Data Security Standard: PCI DSS is the Payment Card Industry global data security standard adopted by the payment card brands for all entities that process, store or transmit cardholders data and/or sensitive authentication data. It consists of steps that mirror security best practices[27].
- Manufacturing guidelines: One of the important standers depends on manufacturing guidelines. Some technical products must rely on the manufacturer's instructions in terms of security because they are not subject to international standards. For this reason, the written instructions from the factory experts should be used in detail[28].
- Best Practice: it represents compliance in successful organizations regarding some of information security users for example British Standard 7799 Part 3, Common Criteria (also known as ISO/IEC 15408), National Information Security Technology Standard Specification, and SANS Security Policy Resource. These procedures are not intended to replace any of the other business frameworks, but to make other f companies frameworks workable usually use NIST, FFIEC, HIPAA, US CERT, ISO and DSS 27001[29].

In his research, Nader obtained the participation of security knowledge sharing for information; teamwork, intrusion, and practice, all have a significant impact on the attitude of employees to comply with organizational information security policies. However, the facility does not have a significant impact on staff attitude towards compliance with information security policy[30].

Information security policy plays a crucial role in the management of information systems security. A security breach can result in loss of revenue and damage to the reputation of the company or institution, as well as excessive fines and legal proceedings. When you begin to develop your security plan, always remember that your network is your weakest link, so you must secure it completely. Previous research has led to the intention of staff compliance with mixed results, which any of the variables investigated are interfering with the severity of the penalties for the intentions of the employees of the ISPs[31].

Regulatory and legal incompatibility and compliance have serious financial implications for international companies, including financial fines, loss of business, repudiation, and class action lawsuits. Ponemon conducted a survey as part of a study that found that the companies that participated in the survey had been subjected to a rate of two attacks per week. In the year 2016, which represents an increase of 23% over the next year, and this increase caused companies to lose \$ 9.5 million annually in the fight against cyber-crime. In the last year alone, more than 4 billion records of data have been tampered worldwide, an increase of 400 percent over the last two years[32, 33].

Governments around the world are applying new EU General Data Protection regulations that require companies to better protect customer information. For example, one of the basic mandatory regulations, which will be in effect from the year 2018. Regulation (GDPR) includes more stringent data protection requirements by different businesses. This mandatory system applies not only to EU countries but also to global business enterprises. It also has to comply with this system if data is collected and used by residents in EU regions. [32]

GEMALTO, the world's leading digital security solutions provider has exposed nearly 10 billion global data records over the past five years to cases of theft, loss or public disclosure, with an average of 5 million data logs per day. The index is a global database that tracks and analyzes data breaches and the type of data affected and how it is accessed, lost or stolen[34].

In many countries, governments have enacted personal data protection (PDP) regulations, for example, Fair Information Practices (FIPs) in the United States, the privacy of personal data in European countries and the Personal Data Protection Act (PDPA2) in Malaysia[1].

These regulations include provisions on how institutions collect, store and use consumer data. Organizations must adhere to regimes whose failure to impose substantial sanctions. Ensuring information security has become one of the top priorities of many organizations [35] because institutional security and compliance with PDP regulations are important factors influencing consumer decision-making[36].

The American Institute of Certified Public Accountants (AICPA) found that compliance with privacy is one of the most important management issues in technology management. Therefore, organizations should pay more attention to how to ensure compliance behavior in all their employees[37].

In South Korea, an employee at Standard Chartered Korea illegally obtained customer data (from 130,000 customers) from the company's network and shared data with insurance vendors. As a result, Standard Chartered's reputation was severely damaged and data was shared with insurance vendors. As a result, due to the share and use of all this information, the reputation of Standard Chartered Bank has been severely damaged. At this level, data breaches can occur because of the malicious and purposeful disclosure of information against benefits, or unintentional disclosure due to negligence[38].

Paradoxically, two case studies of Brechtsen (2007) [39] found a Norwegian bank and a Norwegian IT company that although employees were familiar with data protection work, they did not perform individual data security procedures. Since these investigations are conducted on respondents from different countries, the contradictory results suggest that this is possible[39].

According to a detailed survey conducted by various organizations in Saudi Arabia for a deep look at how differences are addressed and the general focus on external security threats, there were significant threats within the organizations[16].

3.2. Business Process

Corporate enterprise information systems are important assets to provide a competitive advantage in changing sectoral conditions[40], business process continuity and enterprise resource management. In this regard, information security techniques and valuation techniques are used to examine the level of maturity of the institution and to identify the risks and possible solutions to corporate information systems[41].

This should be done through organizational transformation aspects that take into account strategy, technology, process, and people to account for further benefits from the organization[42].

Organizations may want to implement Business Process Management (BPM) as a management tool in conjunction with the implementation/workflow management systems (e.g. activity, etc.) to reduce the cost of management or through migration to free open source software. This can also help you develop customized software solutions that support your strategy of differentiation[40].

Information security standards are transforming their scope into business process security. This is true, since the introduction of information security control in a business-processing environment is

likely to affect the flow of the business process, while redesigning the business process will certainly have security implications[43].

BPM is dedicated to designing, implementing, monitoring and improving business processes. Systems that support the enactment and implementation of processes have been widely used by companies to simplify and automate processes within the organization[44].

It can be helpful first to determine the company's security requirements at hand. Using business processes and security requirements to analyze information security risks, we can find three areas with slightly different objectives[45, 46].

- First, there are approaches that use security requirements to design or verify workflow systems or business processes. Some of these techniques focus on creating safe business processes or processes based on security requirements.
- Second, there are approaches that focus on the analysis and design of information systems (IS) based on business models or organizational models.
- Third, a business risk analysis is proposed with high-level business processes using risk scenarios and safety objectives.

Increased business increases the need for secure business processes where security problems can affect the company's profits and reputation. However, decisions regarding a reasonable level of security in the business environment are usually done in a neutral manner. The use of current business processes allows the use of operational data that is produced during the execution of readily available business processes. Thus, data collection and analysis are easier and faster than continuous monitoring and evaluation. Improving security in parallel with improvement and in accordance with the changing requirements of business operations. Using business processes allows the company to obtain a more accurate set of data needed to assess the cost-effectiveness of security[47].

3.3. Decrease expenses

Gartner revealed that the volume of information security spending will reach \$ 114 billion worldwide in 2018 and the figures No 2 showed an increase in spending of 12.4 percent this year compared to what was paid in 2017. Gartner, an American research organization operating in 100 countries around the world, predicted that the world will spend about \$ 124 billion in 2019[48].

It believes that the information security industry has been affected during the current year with new usage agreements and privacy policies, while the share of privacy protection within the industry will be less than 10 percent. (GDPR) to protect data on \$ 59 billion of global security spending[48].

Users must determine the operating system they will use depending on their economic and financial potential. By choosing less expensive (or free) operating systems and appropriate simulations, IT spending can be reduced significantly[48].

The cost information system plays an important role in every organization in the decision-making process. An important task of management is to ensure that processes and sectors of activity are controlled, not ultimately over costs. Although access to the organization's goals competes with many control systems (production control, quality control and inventory control), the cost information system is important because it monitors the results of others. Detailed cost analysis, cost of production, loss estimation and labor efficiency assessment provide a solid basis for financial control[49].

A large company can explore the means of IT to achieve cost reductions such as IT / automation, cloud computing, virtual data infrastructure, and large data analytics. All of which the company needs to consider before deciding on adopting a particular technology[50].

The security of the company leads to the reduction of the company's expenses by reducing the lack of need for information security experts to identify the gaps. Therefore, reducing the chance of data loss that could result in serious material losses[51].

3.4. Marketing Edge

The application of information security Measures the increase of customer reliability and thus increase in profit as in prepaid systems in banks. The greater the security of information, the greater the number of customers[52].

There are two approaches to assessing the impact of potential risks, one quantitative and the other qualitative. By using quantitative assessment, the expected losses can be translated into financial value, even if the loss is significant[52].

Qualitative assessment is used when we fail to identify the impact of events that are difficult to predict, such as loss of reputation or abuse of the political status of the State or diplomatic embarrassment[52].

The focus is on the risks that will have a negative economic impact on the institution, and it is expected that causing various losses such as operating losses, legal losses, financial losses, strategic losses, and impact on the reputation of the institution[53].

The security of information systems require an economic perspective to balance the costs of information systems security operations with losses that could result from theft, sabotage or denial of service or anything that would tamper with the goals of achieving the secure environment of information systems[53].

The researcher believes that the risk analysis should be measured by cost/benefit analysis, where comparisons must be made between the expected threats and what will cost for what can be offset by the cost of counter-security measures, which represent solutions, but sometimes the organization needs to provide antipersonnel agents at no cost. Panko, agrees in part with what he has put forward and adds that it is worth noting that not all risks can be identified, nor is it to eliminate all hazards and remaining risks called residual risks[54].

Humpert stated that the costs of information systems security fall into four dimensions: development costs, which include research, consulting and the costs of the tools used to protect information systems is considered as direct costs, and non-cost costs include the cost of risks including uncertainties and residual risks[55].



Figure No 2 : The cost dimensions of information systems show direct and indirect costs [12].

4. Research methodology

The researcher relies on the use of the inductive method based on qualitative research. Hence, the researcher starts from the data collection, then observations from previous studies. That is noted to reach certain results (this means that the hypotheses and theories derived from the data during the data collection process) and after analysis. The researcher here examines the data by an expert or specialist to describe any patterns and relations. It is noted here that the construction of the theory according to the entrance to qualitative research is systematically and the data collected through interviews, observations and previous studies, then the theory that will be built is suitable for the field of application.

5. Data analysis and findings

This section present both analysis of collecting data and findings thought the research. The main source of data collection of this paper is through case studies 345 companies from 2004 to 2019[56]. All of them suffered from various types of cyberattacks or data loss due to employees mistake company regard with the big report. As can be seen in figure No 3 the majority of companies (96%) applied any types of compliance, but the attacks are effect on their marketing, changing business process for solve their problems and lost a lot of money.



Figure no 3: Information Security Factors in Organizations

Table 1 display world's biggest data breaches and hacks that implement information security factors in deferent types of organizations. Without a doubt, most of the organizations tried to apply GRDP or ISO 270001 in last years. On the other hand, an attack force on organizations marketing with their customers. At the same time, the organizations after suffered of risk trying to change their process. Consequently, it becomes necessary to propose an information security awareness program.

	Compliance Applied	Marketing Edge	Decrease Expenses	Business Process	No. of SECTOR
Academic	11	10	8	7	11
Energy	3	3	3	2	3
Financial	35	29	25	28	35
Gaming	11	8	8	6	11

Government	38	32	30	26	38
Healthcare	40	38	34	37	43
Tech	13	11	10	10	13
Transport	9	6	6	9	9
Telecoms	14	12	10	12	14
Others	159	158	130	135	168

Table1: Information Security Factors in 345 companies.

6. Results

Some organizations in the world have implemented a policy of compliance with their own information security to protect consumer information, increase market value and provide ideal information security. The literature review shows that many organizations are either unaware of or inclined to ignore policy, thereby increasing the risk of non-compliance. To assist instructions or organizations in managing compliance among their employees, we used the factors influencing administrative decision-making by understanding these factors and how to apply them correctly. Effective and targeted strategies can be developed to educate organizations accordingly and to promote and enhance information security among organizations. The Security policy should be reviewed independently once a year or earlier if circumstances warrant. The following figure 3 illustrates the factors affecting administrative decision making.



Figure 4: The impact of administrative decision-making

After building the model factors affecting the decision-making of the Iraqi organizations by the researcher, the results of this study were presented to engineer Hamza Adnan Dawod, a data security expert who works for International Smart Card Company (ISC- QICARD) as an Information security management system project manager and responsible for data security in the company. The points identified in previous studies were explained and five questions identified. Below are the questions and answers that were conducted by the researcher to the news in order to evaluate the results of the research.

1. How compliance policy is adopted in Iraqi organizations? Certainly the policy of compliance in the organizations must be adopted, because Iraqi organizations need legislation to deter use of bad technology.
2. Is Information Security Affecting Workflow in your company? It is clear that companies in the world are emphasizing the security of data as an important factor in the course of business. Companies avoid many problems as a result of security.

3. Does the policy of proverbs reduce the amounts paid for data security? Yes of course, because compliance application reduces security problems. On the other hand, companies that have not implemented this policy are forced every time they are exposed to a security breach or attacks to the use of an expert or a security company data and this costs the company time, effort and money.
4. Does the use of information security affect product marketing? Today, especially in companies dealing with money such as banks and prepaid card companies, they need to raise the level of information security to increase customer confidence in the product.
5. Finally, do you think that the factors that have been challenged in advance can affect the decision-making process in the organization? All of these factors are effective, and all these factors must be taken into account in the decision-making process.

7. Conclusion and future work

This study proves making the administrative decision is essential for the government in general. However, government decision-making procedures have been inefficient in the past. Based on case studies of the organizations, information security factors model is proposed in this study. Thus improving the effectiveness and efficiency of decision-making through four factors compliance, business process, decrease expenses and marketing edge. As E-government develops increasingly, organizations consider these factors, apply them to improve their management decision, and may act the process of administrative decision making directly affects and efficiency of public administration. In the future, the researcher proposes that Iraqi organizations should prepare action steps to implement the compliance policy and to move to electronic governance through the application of the GDPR measurement.

References

- [1] H. N. Chua, S. F. Wong, Y. C. Low, and Y. Chang, "Impact of Employees' Demographic Characteristics on the Awareness and Compliance of Information Security Policy in Organizations," *Telematics and Informatics*, 2018.
- [2] I. Mann, *Hacking the human: social engineering techniques and security countermeasures*: Routledge, 2017.
- [3] M. Ojala, M. Pantti, and J. Kangas, "Professional role enactment amid information warfare: War correspondents tweeting on the Ukraine conflict," *Journalism*, vol. 19, pp. 297-313, 2018.
- [4] J. S. Brown and P. Duguid, *The Social Life of Information: Updated, with a New Preface*: Harvard Business Review Press, 2017.
- [5] C. Avgerou and G. Walsham, *Information Technology in Context: Studies from the Perspective of Developing Countries: Studies from the Perspective of Developing Countries*: Routledge, 2017.
- [6] M. P. Thompson, G. Walsham, and I. Faik, "Designing for ICT-enabled openness in bureaucratic organizations: problematizing, shifting and augmenting boundary work," 2018.
- [7] M. H. Rehmani, M. Reisslein, A. Rachedi, M. Erol-Kantarci, and M. Radenkovic, "Guest editorial special section on smart grid and renewable energy resources: Information and communication technologies with industry perspective," *IEEE Transactions on Industrial Informatics*, vol. 13, pp. 3119-3123, 2017.
- [8] P. Attfield, M. Doyle, and V. Ting, "Secure Data Protection and Encryption Techniques for Computing Devices and Information Storage," ed: Google Patents, 2018.
- [9] I.-D. Barbu, C. Pascariu, I. C. Bacivarov, S.-D. Axinte, and M. Firoiu, "Intruder monitoring system for local networks using Python," in *Electronics, Computers and Artificial Intelligence (ECAI), 2017 9th International Conference on*, 2017, pp. 1-4.

- [10] Y. Kawanishi, H. Nishihara, D. Souma, and H. Yoshida, "Detailed Analysis of Security Evaluation of Automotive Systems Based on JASO TP15002," in *International Conference on Computer Safety, Reliability, and Security*, 2017, pp. 211-224.
- [11] W. Stallings, *Cryptography and network security: principles and practice*: Pearson Upper Saddle River, NJ, 2017.
- [12] J. Garmany, J. Walker, and T. Clark, *Logical database design principles*: CRC Press, 2005.
- [13] W. Kim, "Cloud computing: Today and tomorrow," *Journal of object technology*, vol. 8, pp. 65-72, 2009.
- [14] C. A. Lengnick-Hall and T. E. Beck, "Adaptive fit versus robust transformation: How organizations respond to environmental change," *Journal of Management*, vol. 31, pp. 738-757, 2005.
- [15] M. Herald. (20 September 2018). *Gartner Survey Finds Consumers Would Use AI to Save Time and Money*. Available: <https://www.menaherald.com/en/tech/information-technology/gartner-survey-finds-consumers-would-use-ai-save-time-and-money>
- [16] M. Alsaif, N. Aljaafari, and A. R. Khan, "Information Security Management in Saudi Arabian Organizations," *Procedia Computer Science*, vol. 56, pp. 213-216, 2015.
- [17] M. E. Whitman and H. J. Mattord, *Principles of information security*: Cengage Learning, 2011.
- [18] F. H. Alqahtani, "Developing an Information Security Policy: A Case Study Approach," *Procedia Computer Science*, vol. 124, pp. 691-697, 2017.
- [19] I. Hwang and O. Cha, "Examining technostress creators and role stress as potential threats to employees' information security compliance," *Computers in Human Behavior*, vol. 81, pp. 282-293, 2018.
- [20] K. Newstadt and T. Brown, "Dynamic endpoint compliance policy configuration," ed: Google Patents, 2010.
- [21] E. Myler and G. Broadbent, "ISO 17799: standard for security: organizations can use ISO 17799 as a model for creating information security policies and procedures, assigning roles and responsibilities, documenting operational procedures, preparing for incident and business continuity management, and complying with legal requirements and audit controls," *Information Management Journal*, vol. 40, pp. 43-51, 2006.
- [22] A. Kolah, *The GDPR Handbook: A Guide to Implementing the EU General Data Protection Regulation*: Kogan Page Publishers, 2018.
- [23] V. Stanciu and S.-M. Rîndașu, "The Impact of General Data Protection Regulation in The Accounting Profession—Evidences from Romania," 2018.
- [24] J. Arturo and T. Romero, "PDCA-based approach to Environmental Value Stream Mapping (E-VSM)," *Journal of cleaner production*, 2018.
- [25] M. Khouja, I. B. Rodriguez, Y. B. Halima, and S. Moalla, "IT Governance in Higher Education Institutions: A Systematic Literature Review," *International Journal of Human Capital and Information Technology Professionals (IJHCITP)*, vol. 9, pp. 52-67, 2018.
- [26] S. Taylor, "ITIL V3 Foundation Complete Certification Guidebook: ITIL V3 Study Guide to Pass In First Attempt," 2018.
- [27] A. Nagar and K. P. Joshi, "A Semantically Rich Knowledge Representation of PCI DSS for Cloud Services," in *6th International IBM Cloud Academy Conference ICACON 2018, Japan*, 2018.
- [28] A. Singla, I. S. Ahuja, and A. Sethi, "Technology push and demand pull practices for achieving sustainable development in manufacturing industries," *Journal of Manufacturing Technology Management*, vol. 29, pp. 240-272, 2018.
- [29] Y. Kawanishi, H. Nishihara, D. Souma, H. Yoshida, and Y. Hata, "A Study on Quantitative Risk Assessment Methods in Security Design for Industrial Control Systems," in *2018 IEEE 16th Intl Conf on Dependable, Autonomic and Secure Computing, 16th Intl Conf on Pervasive Intelligence and Computing, 4th Intl Conf on Big Data Intelligence and Computing and Cyber Science and Technology Congress (DASC/PiCom/DataCom/CyberSciTech)*, 2018, pp. 62-69.
- [30] N. S. Safa, R. V. Solms, and S. Furnell, "Information security policy compliance model in organizations," *computers & security*, 2016.

- [31] X. Chen, D. Wu, L. Chen, and J. K. Teng, "Sanction severity and employees' information security policy compliance: Investigating mediating, moderating, and control variables," *Information & Management*, 2018.
- [32] P. Institute, "Cost of Cyber Crime Study & the Risk of Business Innovation " October 2016.
- [33] R. B. Security, "The 2016 Year End Data Breach Quick View," 2017.
- [34] J. P. DiMaggio and E. N. Stone, "Determining risk level and maturity of compliance activities," ed: Google Patents, 2018.
- [35] S. Ransbotham and S. Mitra, "Choice and chance: A conceptual model of paths to information security compromise," *Information Systems Research*, vol. 20, pp. 121-139, 2009.
- [36] H. Xu, X. R. Luo, J. M. Carroll, and M. B. Rosson, "The personalization privacy paradox: An exploratory study of decision making process for location-aware marketing," *Decision support systems*, vol. 51, pp. 42-52, 2011.
- [37] M. Warkentin, A. C. Johnston, and J. Shropshire, "The influence of the informal social learning environment on information privacy policy compliance efficacy and intention," *European Journal of Information Systems*, vol. 20, pp. 267-284, 2011.
- [38] T. Herath and H. Raghav Rao, "Control mechanisms in information security: a principal agent perspective," *International Journal of Business Governance and Ethics*, vol. 5, pp. 2-13, 2009.
- [39] E. Albrechtsen, "A qualitative study of users' view on information security," *Computers & security*, vol. 26, pp. 276-289, 2007.
- [40] Ş. Eroğlu and T. Çakmak, "Information as an organizational asset: assessment of a public organization's capabilities in Turkey," *Information Development*, p. 0266666918811004, 2018.
- [41] A. Kumar, *Business process management*: Routledge, 2018.
- [42] L. G. Pee and A. Kankanhalli, "A model of organisational knowledge management maturity based on people, process, and technology," *Journal of Information & Knowledge Management*, vol. 8, pp. 79-99, 2009.
- [43] G. Wangen and E. A. Snekenes, "A comparison between business process management and information security management," in *Computer Science and Information Systems (FedCSIS), 2014 Federated Conference on*, 2014, pp. 901-910.
- [44] J. Mendling, I. Weber, W. V. D. Aalst, J. V. Brocke, C. Cabanillas, F. Daniel, *et al.*, "Blockchains for business process management-challenges and opportunities," *ACM Transactions on Management Information Systems (TMIS)*, vol. 9, p. 4, 2018.
- [45] S. Taubenberger and J. Jürjens, "IT security risk analysis based on business process models enhanced with security requirements," in *Proceedings of the First International Modeling Security Workshop (MODSEC'08), The Eleventh International Conference on Model Driven Engineering Languages and Systems (MODELS'08)*, <http://ceur-ws.org>, 2008.
- [46] P. Kobezak, R. Marchany, D. Raymond, and J. Tront, "Host Inventory Controls and Systems Survey: Evaluating the CIS Critical Security Control One in Higher Education Networks," in *Proceedings of the 51st Hawaii International Conference on System Sciences*, 2018.
- [47] T. Neubauer, M. Klemen, and S. Biffl, "" Business process-based valuation of IT-Security"; Vortrag: 7th International Workshop on Economics-Driven Software Engineering Research EDSER'05, St. Louis, Missouri; 15.05. 2005; in:" Proceedings of the 7th International Workshop on Economics-Driven Software Engineering Research EDSER'05, Volume 30 Issue 4", ACM Press,(2005), ISBN: 1-59593-118-x; S. 1-5."
- [48] A. Skendzic, B. Kovacic, and I. Jugo, "Decreasing information technology expenses by using emulators on Windows and Linux platforms," in *MIPRO*, 2011, pp. 1387-1390.
- [49] G. V. Lepadatu, "The Importance of The Cost Information In Making Decisions," *Metalurgia International*, vol. 17, p. 123, 2012.
- [50] H. P. Fung, "Using porter five forces and technology acceptance model to predict cloud computing adoption among IT outsourcing service providers," 2014.
- [51] A. Aljabre, "Cloud computing for increased business value," *International Journal of Business and social science*, vol. 3, 2012.
- [52] M. Weber, "The business case for corporate social responsibility: A company-level measurement approach for CSR," *European Management Journal*, vol. 26, pp. 247-261, 2008.

- [53] C. G. Giske, D. L. Monnet, O. Cars, and Y. Carmeli, "Clinical and economic impact of common multidrug-resistant gram-negative bacilli," *Antimicrobial agents and chemotherapy*, vol. 52, pp. 813-821, 2008.
- [54] M. Altman, J. Gill, and M. P. McDonald, *Numerical issues in statistical computing for the social scientist* vol. 508: John Wiley & Sons, 2004.
- [55] A. Bierhaus, K.-M. Haslbeck, P. M. Humpert, B. Liliensiek, T. Dehmer, M. Morcos, *et al.*, "Loss of pain perception in diabetes is dependent on a receptor of the immunoglobulin superfamily," *The Journal of clinical investigation*, vol. 114, pp. 1741-1751, 2004.
- [56] D. McCandless, "World's biggest data breaches & hacks," *Information is Beautiful*, 2019.